# Critical Appraisal of Scientific Papers

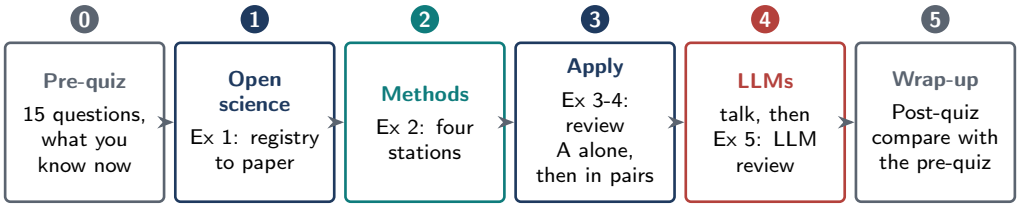
## Reading a clinical paper the way a good peer reviewer does, with and without AI

Imad El Badisy, PhD

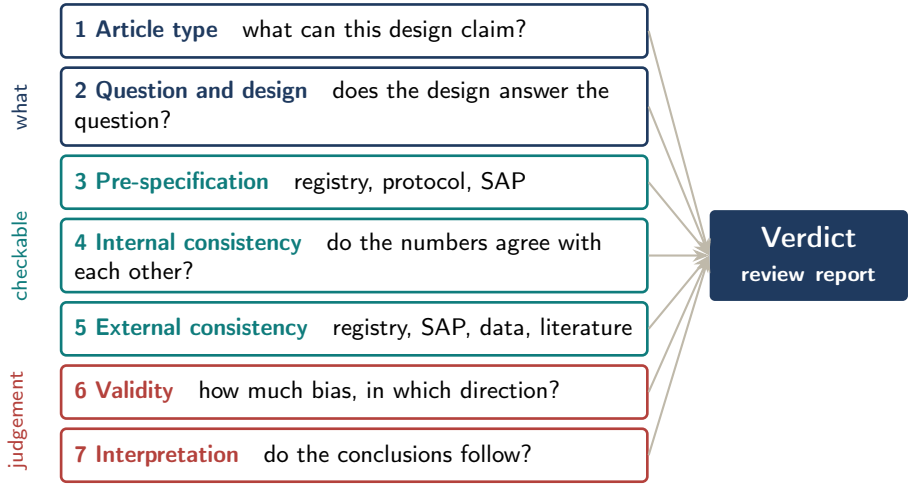
CM6RI

One-day workshop

# The day: learn on two real papers, apply on one



# One tool all day: seven layers, then a verdict




**Internal consistency**: the numbers in the paper agree with each other. **External consistency**: the paper agrees with its registry, protocol, SAP and other reports. **SAP**: statistical analysis plan.

Checklist: handouts/checklist-7-layers.md

# Pre-quiz

individual

10

minutes

## START NOW

**Instructions and answer grid:** pack p. 2-3

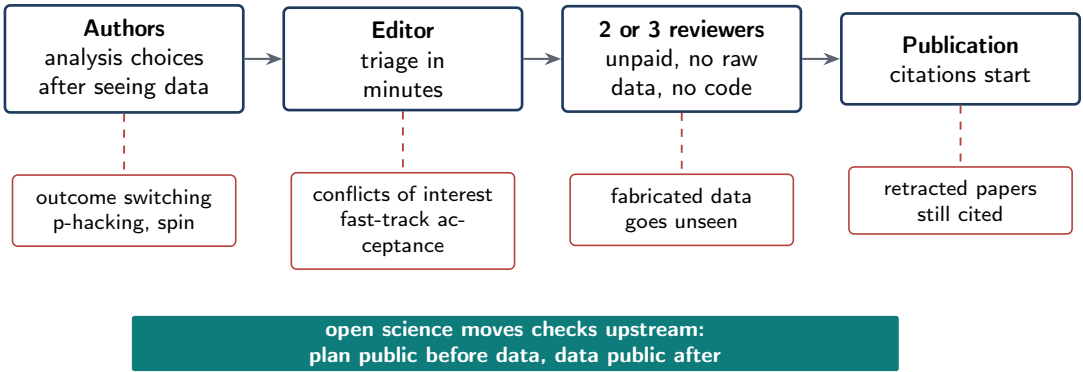
15 questions, one answer each.

**STOP when time is up → next: block 1, open science**

# 1 Open science and pre-specification

What was promised, what was published, and who can check. Layers 1, 2, 3 and 5.

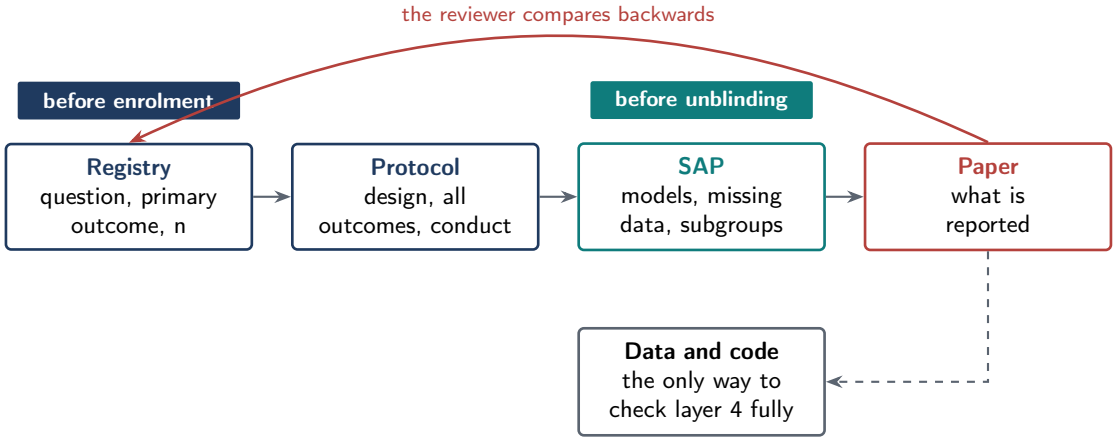
# Where publishing breaks




**Outcome switching**: reporting a different primary outcome from the one registered. **p-hacking**: trying analyses until one gives $p < 0.05$. **Spin**: wording that makes results look better than they are. **Fast-track**: acceptance after a shortened review (Gautret: accepted the day after submission).

Cases used today: Gautret 2020 (fast-track, retracted 2024); Elgazzar 2020 (withdrawn, fabricated records).

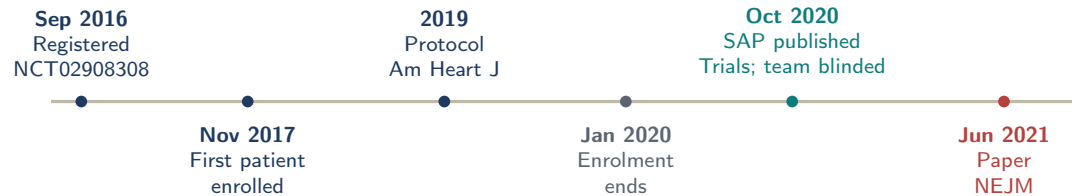
# Four documents, each locks something before the data are seen




**Registry**: public record of a study (e.g. ClinicalTrials.gov). **SAP**: statistical analysis plan, the analyses written in detail before the data are seen. **Unblinding**: the moment the team learns who received which treatment.

SPIRIT 2025 (protocols), CONSORT 2025 (reports), ICH E9(R1) (estimands and SAP).

# TTM2: a paper trail you can follow



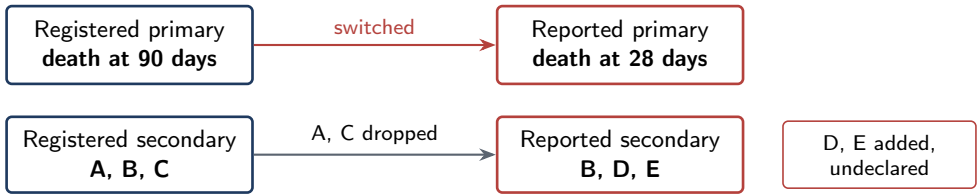

**Primary outcome**
death at 6 months:
same in all four documents

**Sample size 1200 to 1900**
for TAME co-enrolment,
before any results

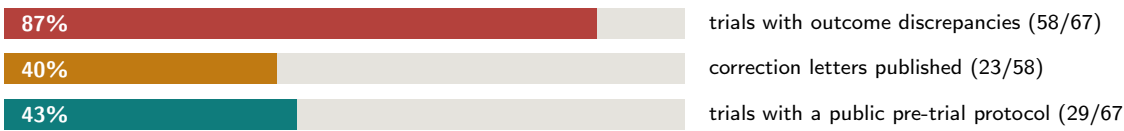
**One detail**
SAP relative reduction 13.6%,
paper says "15%"

**TAME co-enrolment**: patients could also enter the TAME trial; the sample size was raised so both trials kept their power. **Relative reduction**: (control risk − treated risk) / control risk.

# Outcome switching: the registered outcome quietly becomes another



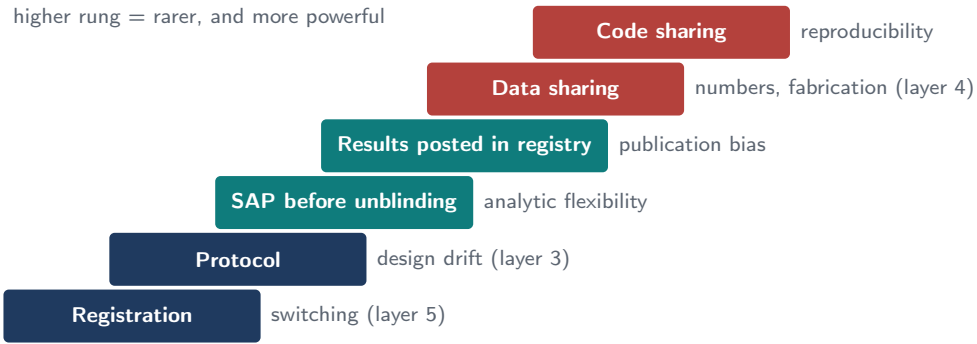

**COMPare: all 67 trials over 6 weeks in NEJM, Lancet, JAMA, BMJ, Annals**




**Discrepancy** (COMPare): a registered outcome not reported, or a new outcome added without saying so. **Correction letter**: a letter to the journal pointing out the discrepancy.

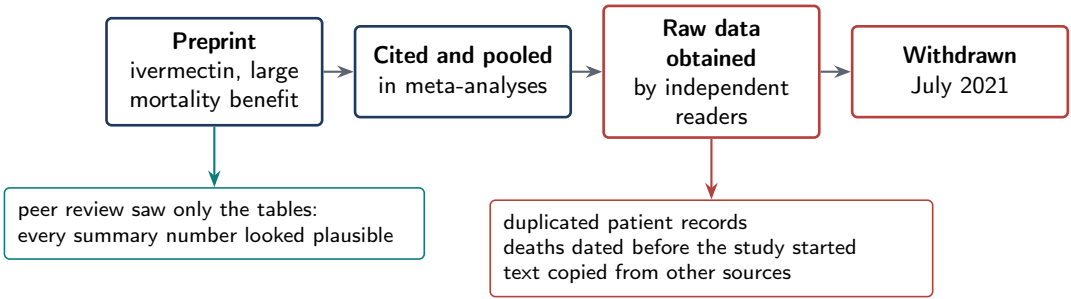
Goldacre et al. COMPare. Trials 2019;20:118 (materials/references/method-goldacre-compare-2019.pdf).

# Open science is a ladder: each rung makes one more check possible




**Analytic flexibility**: many defensible analyses exist; choosing after seeing the data favours the one that "works". **Publication bias**: negative results are published less often. **Reproducibility**: same data and code give the same numbers.

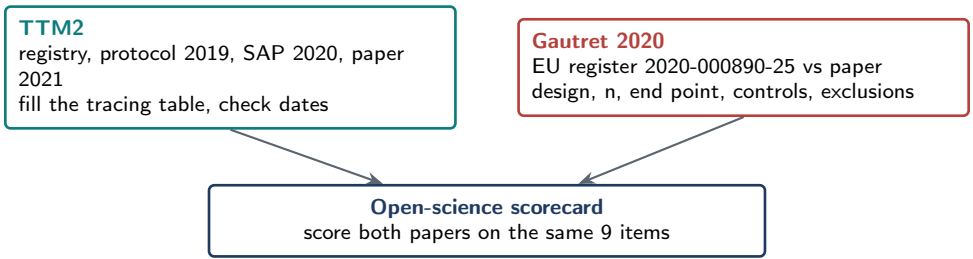
# Elgazzar 2020: only the raw data could show it




**Preprint**: a manuscript posted online before peer review. **Meta-analysis**: a pooled estimate from several studies; one fabricated study can shift it. **Raw data**: one row per patient, before any summary.

Research Square withdrawal note, 14 July 2021 (materials/references/article-elgazzar-2020-withdrawn.pdf).

# Exercise 1 Trace two papers from promise to publication




**Pairs 50 minutes + 10 debrief pack p. 4-5**

Part A TTM2 (25 min), Part B Gautret (20 min), Part C scorecard (5 min).

## Exercise 1 — Registry to paper

pairs, laptops open

50

minutes

### START NOW

**Instructions and answer grid:** pack p. 4-5

Part A (25 min): fill the TTM2 table, answer Q1-3.
Part B (20 min): fill the Gautret table, answer Q4-6.
Part C (5 min): score both papers, answer Q7.
TTM2 protocol and SAP: materials/workshop/

**STOP when time is up → next: debrief (10 min); break**

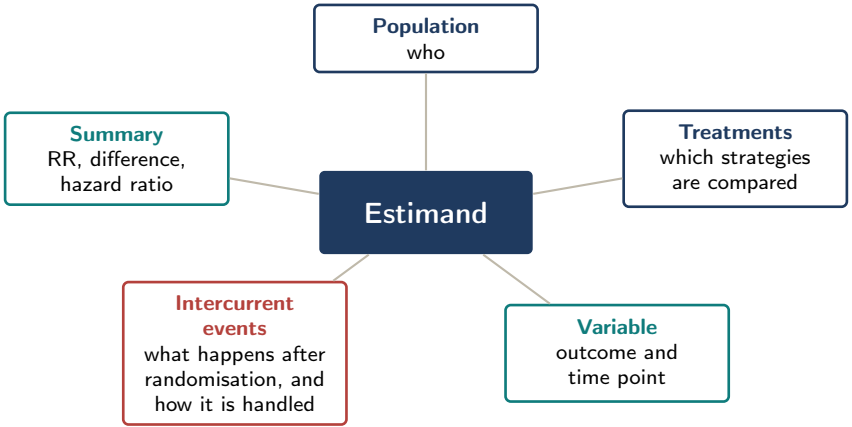
## Break — 15 minutes

Next: block 2, statistical methodology on real papers

# 2 Statistical methodology on real papers

Estimand, confidence intervals, forensics, bias, spin. Layers 2, 4, 6 and 7, on TTM2 and Gautret.

# The estimand: five attributes that define the question




**Estimand**: the exact quantity the study sets out to estimate. **Intercurrent event**: something after randomisation that changes how the outcome can be read (death, stopping treatment, rescue therapy). **Summary measure**: the number that compares the groups.

ICH E9(R1) addendum on estimands, 2019.

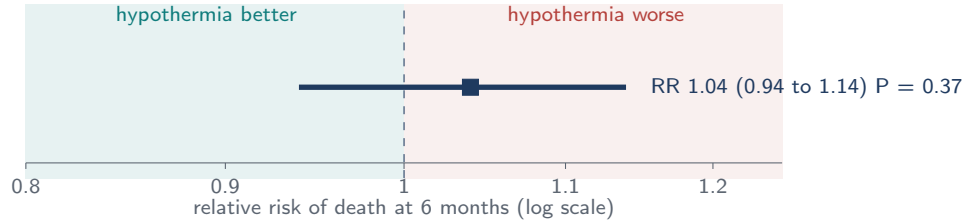
# Same words, different questions: TTM2 vs Gautret

| | TTM2 | Gautret |
|---|---|---|
| **Population** | comatose adults after OHCA | inpatients, one centre; controls from other centres |
| **Treatments** | 33 °C vs normothermia | HCQ (± AZ) vs no HCQ, not assigned |
| **Variable** | death by 6 months | PCR negative at day 6 (surrogate) |
| **Intercurrent** | treatment policy | 6 treated excluded: ICU, death, discharge, stop |
| **Summary** | relative risk | proportions, many p-values |

**OHCA**: out-of-hospital cardiac arrest. **Treatment policy**: the outcome counts whatever happens after randomisation (as in intention to treat). **Surrogate**: a biomarker (here a PCR result) used in place of a clinical outcome. **HCQ, AZ**: hydroxychloroquine, azithromycin.

Dankiewicz et al. NEJM 2021; Gautret et al. Int J Antimicrob Agents 2020 (retracted 2024).

# Read the interval, not the p-value



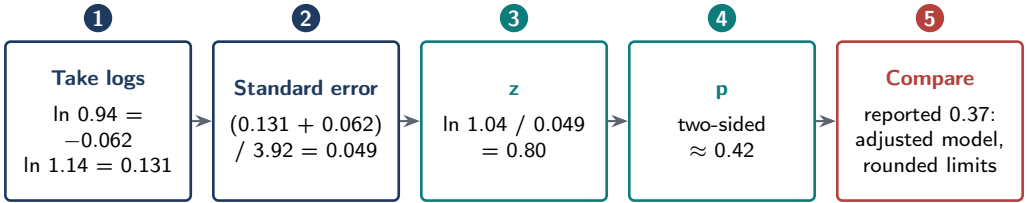

**Correct**
no difference shown; effects from a 6% reduction to a 14% increase remain compatible

**Wrong**
"hypothermia does not work"
"the two treatments are equivalent"

**RR**: relative risk = risk in the treated group / risk in the control group; 1 = no difference. **95% CI**: the range of effects compatible with the data. **Equivalence**: needs a margin fixed in advance and a CI inside it; a non-significant p does not show it.

TTM2: death 465/925 (50%) vs 446/925 (48%).

# From a confidence interval back to a p-value




**Rule of thumb.** A 95% CI that excludes the null implies $p < 0.05$; one that includes it implies $p > 0.05$. A paper that breaks this rule needs a question to the authors.

**Logs**: ratios (RR, OR, HR) are symmetric on the log scale, so the calculation is done there. **SE**: standard error, the precision of the estimate; a 95% CI is $\pm 1.96$ SE, so its width is 3.92 SE. **z**: estimate / SE; $|z| > 1.96$ means $p < 0.05$.

Altman and Bland, BMJ 2011;343:d2090. Toolkit, layer 4.

# Forensics: four checks you can do with a phone

**Percentages**

recompute every $n/N$ you can find

87/120 = 72.5%, not 75.5%

**CI vs p**

do they agree?

OR 1.32 (1.01 to 1.73) with p = 0.11: no

**GRIM**

mean × n must be close to an integer

n = 13, mean 2.41: 31.33: impossible

**Flow**

every patient from enrolled to analysed

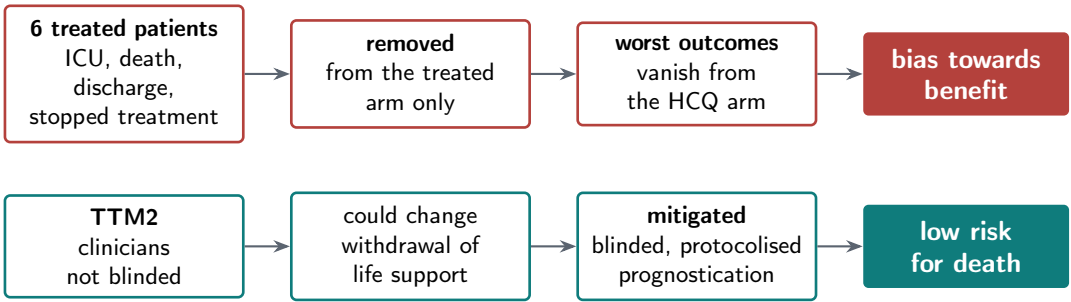
TTM2 adds up; Gautret: no flow diagram

**an inconsistency is a question to the authors, not an accusation**

**GRIM** (granularity-related inconsistency of means): with whole-number data, mean × n must give a whole number, allowing for rounding; works for small n. **Flow**: count every patient from enrolled to analysed in each arm; numbers must add up and every loss be explained (CONSORT flow diagram). **OR**: odds ratio.

Brown and Heathers, Soc Psychol Personal Sci 2017 (GRIM); Nuijten et al., Behav Res Methods 2016 (statcheck).

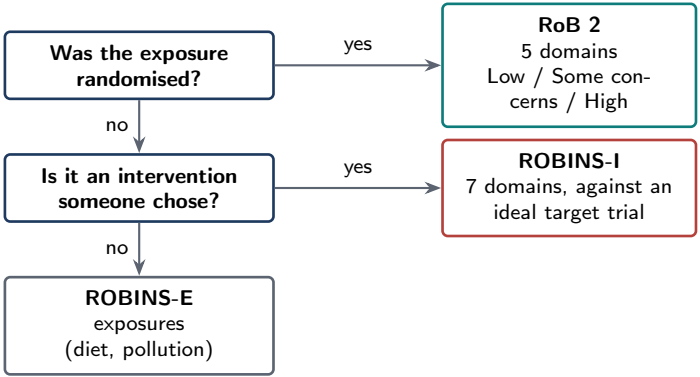
# Bias has a direction: ask which way it pushes




**Per protocol**: analysing only patients who followed the protocol. **Outcome-dependent exclusion**: removing patients because of what happened to them, here the worst outcomes. **Blinded prognostication**: neurological prognosis assessed by someone who did not know the arm.

Calling outcome-dependent exclusions "per protocol" does not remove the bias.

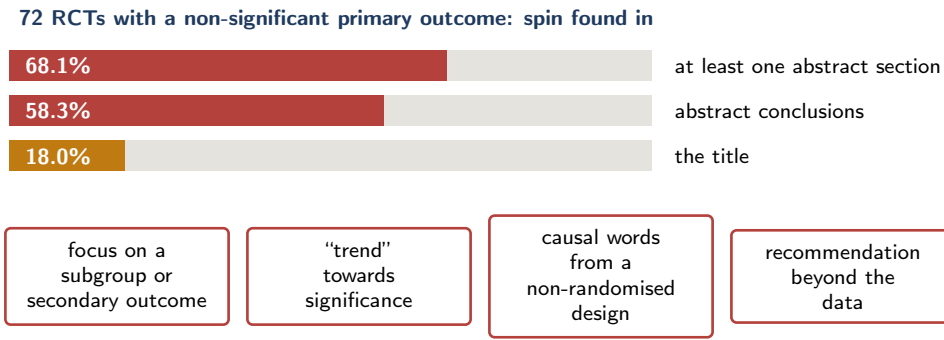
# Which risk-of-bias tool?




**RoB 2**: risk-of-bias tool for randomised trials. **ROBINS-I / -E**: for non-randomised studies of interventions / exposures. **Target trial**: the ideal randomised trial the study tries to imitate. **Confounding**: a factor linked to both treatment and outcome.

Sterne et al. BMJ 2019 (RoB 2); BMJ 2016 (ROBINS-I); Higgins et al. Environ Int 2024 (ROBINS-E).

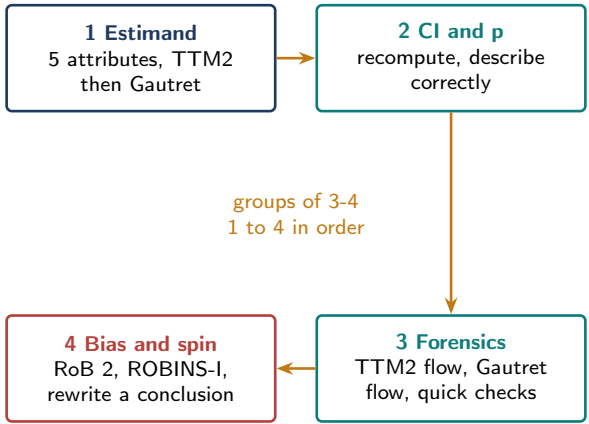
# Spin: the abstract says more than the results




**Spin**: reporting that makes a treatment look better than the results show. **Non-significant primary outcome**: $p \geq 0.05$ on the main pre-specified outcome. **Trend**: wording for a non-significant result presented as almost a benefit.

Boutron et al. JAMA 2010;303:2058.

# Exercise 2 Four stations, 15 minutes each




Pack p. 6-7

## Exercise 2 Four stations — groups of 3-4

$4 \times 15$

**minutes**

START NOW

**Instructions and answer grid:** pack p. 6-7

All groups, in order: $1 \to 2 \to 3 \to 4$.
About 15 minutes per station; I call each change.
Formulas: toolkit, p. 23-25

**STOP when time is up → next: debrief (10 min); lunch**

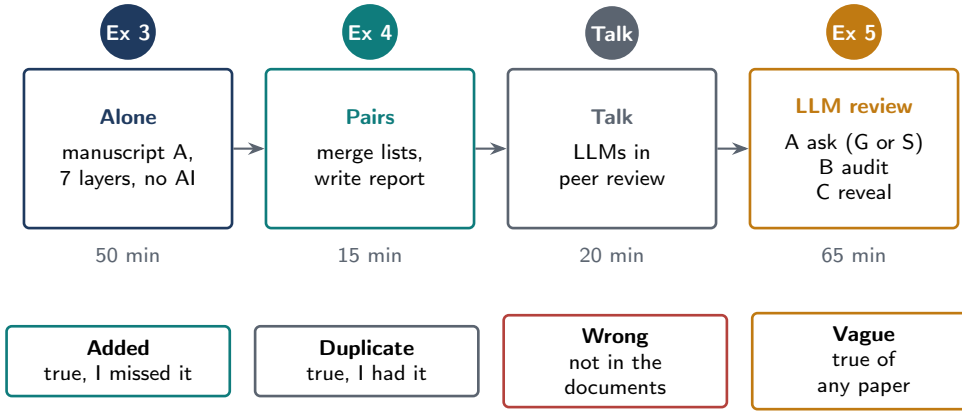
## Lunch — 60 minutes

Next: block 3, apply the method to manuscript A

# 3 Apply the method, then ask an LLM

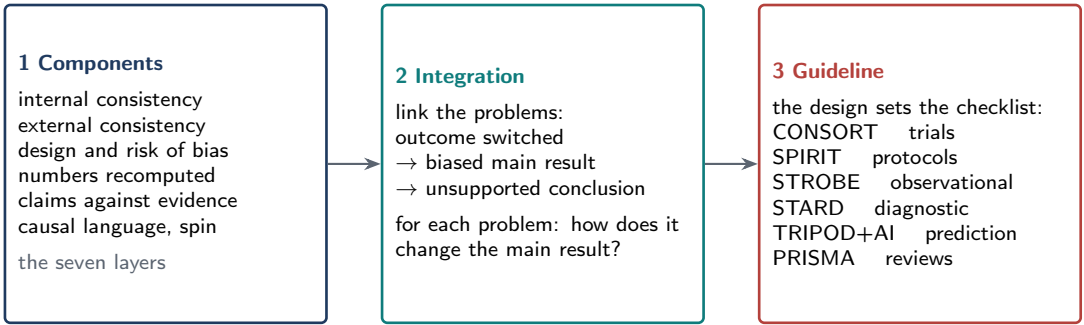
One new paper, reviewed alone with the seven layers, then in pairs.

# How the afternoon works




Exercises 3 to 5, pack p. 8-18; llm-exercise/manuscript-A.md. Manuscript A is fictional: never paste a manuscript you review for a journal.

# A review is a process: components, integration, guideline




**Reporting guidelines**: CONSORT randomised trials; SPIRIT trial protocols; STROBE observational studies; STARD diagnostic accuracy; TRIPOD+AI prediction models, including machine learning; PRISMA systematic reviews.

The same mental model is what you give an LLM in prompt S. Links: handouts/toolkit.md, layer 1.

# Exercise 3 Review manuscript A

alone, no AI

50

minutes

## START NOW

**Instructions and answer grid:** pack p. 8

Read manuscript A (p. 9-12) with the checklist (p. 19-22), layer by layer.
Fill the table on p. 8: one row per problem, with location, what is wrong (show your arithmetic), why it matters.

**STOP when time is up → next: Exercise 4, pairs**

# The review report: what an author can act on

**Summary** 3 to 5 neutral sentences: you understood the paper

**Major issues** at most four, each with three parts → **1 the problem** **2 why it matters** **3 what to do**

**Minor issues** wording, tables, references

**Recommendation** and your confidence in the main conclusion

**Confidential to the editor** concerns you cannot prove

Not: "The statistics are poor." Instead: "Six treated patients are excluded after ICU transfer or death (Section 3.1). Because they are the worst outcomes, this biases the result towards benefit. Please report all 26 treated patients."

# Exercise 4 Pair report

pairs

15

minutes

## START NOW

**Instructions and answer grid:** pack p. 13

Merge your two Exercise 3 tables: what did one of you see that the other missed?
Write the report in the template (p. 14-15): summary, major issues, minor issues, recommendation.

**STOP when time is up → next: break, then the talk on LLMs**

## Break — 15 minutes

Next: block 4, LLMs in peer review, then Exercise 5

# 4 LLMs in peer review: practical concerns

Policies, confidentiality, invented facts, and what the evidence shows. Then Exercise 5: an LLM reviews manuscript A, you audit it, and the reveal.

# Where an LLM could enter, and where it must not

**Your own paper**
before submission:
check reporting

**Published papers**
screening, extraction
(verified)

**allowed, with verification and disclosure**

**Confidential manuscript**
you review for
a journal

**do not upload**

**Your written report**
language polishing

**only if policy allows**

**NIH (2023)**
reviewers may not use
generative AI on applications

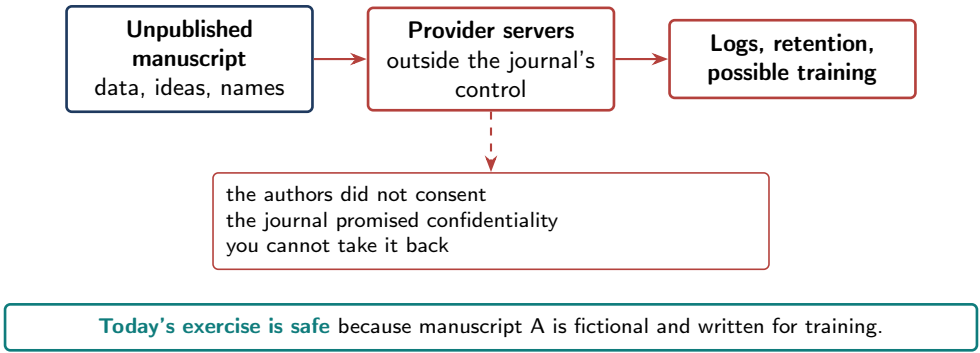
**ICMJE**
no upload where confidentiality
cannot be assured; disclose use

**Most publishers**
reviewers must not upload
manuscripts to AI tools

**LLM**: large language model (ChatGPT, Claude, Gemini, Copilot). **ICMJE**: International Committee of Medical Journal Editors. **Generative AI**: AI that produces text or images.

NIH notice NOT-OD-23-149 (2023); ICMJE Recommendations (updated 2024); check your journal's current policy before every review.

# Confidentiality: what happens when you paste a manuscript




**Retention**: the provider keeps your text for a period. **Training**: the provider may use it to improve future models, so it can resurface.

# Four ways an LLM review goes wrong

**Invented facts**
registry content, protocol details it never saw

**Invented refs**
plausible authors, journals, years; no such paper

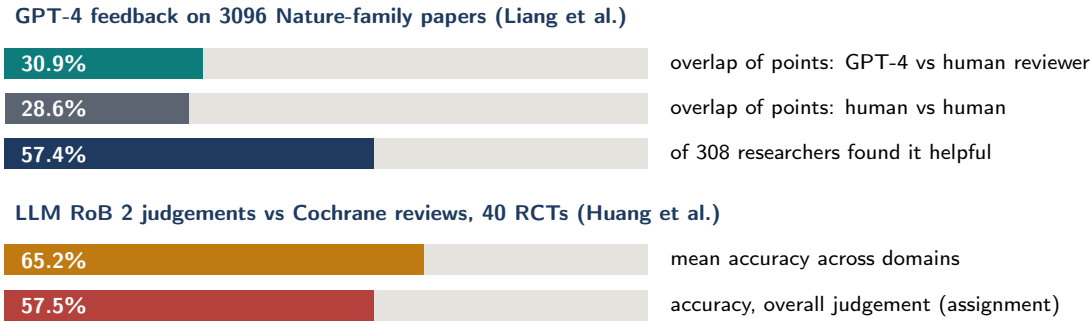
**Wrong arithmetic**
percentages, p from CI, GRIM, stated confidently

**Boilerplate**
"consider confounding" for any paper

**every LLM claim is a hypothesis you must check in the document**

**Invented facts** ("hallucinations"): fluent, confident statements with no source. **Boilerplate**: generic comments true of any paper.

# What the evidence shows so far




**Overlap**: share of the points raised by one reviewer that the other also raised. **Accuracy**: share of LLM judgements that match the Cochrane review.

Liang et al. NEJM AI 2024; Huang et al. J Med Internet Res 2025;27:e70450 (materials/references/). Useful as a second reader; not a replacement.

# When AI conducts research: reproducible is not valid

**PNAS 2026: 288 researchers randomised in 103 teams to reproduce published results**




**An LLM agent on a survival analysis**

right analysis plan in 9 of 9 runs, yet systematic code errors repeated across runs; 8 of 17 summaries fully right, 2 clinically wrong.

**Three levels**

same numbers (reproducible)
$\neq$ right methods (statistically valid)
$\neq$ claim supported (scientifically valid).
Code that runs is only the first level.

**AI-assisted**: humans lead, AI helps. **AI-led**: AI does the work with little human checking. **LLM agent**: an LLM that writes and runs code by itself.

Brodeur et al. PNAS 2026 (social-science reproduction only); Wu et al. J Med Internet Res 2026 (doi:10.2196/99597).

# Two prompts, same paper

**Prompt G (generic), odd seats**

You are a peer reviewer for a general medical journal. Review the paper below.

**Prompt S (structured), even seats**

Work through these components in order; each feeds the next:

1 article type and its reporting guideline
2 external consistency with every source given
3 internal consistency: recompute every number
4 design and risk of bias
5 claims and language
6 integration: how each problem changes the conclusion

Location, what is wrong (show arithmetic), why it matters. Rank. Invent nothing.

**Both**

Paste from "Document 1" to the end. Note the model and its version.

**Prompt**: the instruction you give the LLM. **Generic**: one sentence, no method. **Structured**: the method of this morning, step by step.

Full text of both prompts: Exercise 5, Part A, pack p. 16.

# Exercise 5 LLM review

individual, then plenary

65

minutes

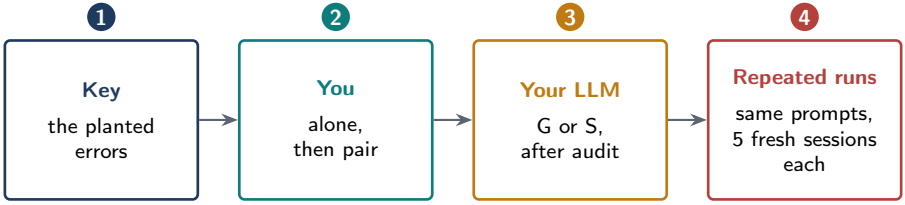
## START NOW

**Instructions and answer grid:** pack p. 16-18

Part A (10 min): odd seat prompt G, even seat prompt S; paste it, then the file llm-exercise/manuscript-A.md from "Document 1".
Part B (20 min): check every LLM point yourself; fill the audit table.
Part C (35 min): reveal with the annotated manuscript A; score table, questions 1-4, your one rule.

**STOP when time is up → next: wrap-up and post-quiz**

# Exercise 5, Part C  Reveal and compare




| | manuscript only | + registry and SAP |
|---|---|---|
| **prompt G** | ? | ? (half the room) |
| **prompt S** | ? | ? (half the room) |

**Planted errors**: errors written into manuscript A on purpose. **Wrong claim**: an LLM point not supported by the documents. **Trap**: a suspicion that only the registry or SAP can confirm.

Errors found, wrong claims, traps flagged. What changes with the prompt? What changes with the sources?

# What repeated runs teach, for any paper

**Documents beat prompts**

a check that needs the registry or SAP cannot be done without it, whatever the prompt

**Structure raises the right questions**

the structured prompt asked what it could not verify; the generic one mostly did not

**Arithmetic is cheap, not free**

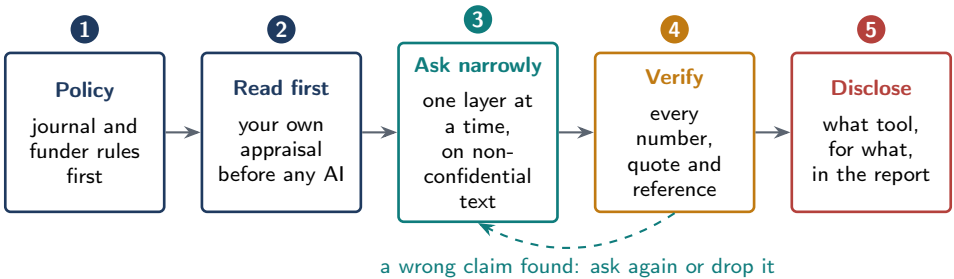
recomputation was near ceiling, yet some runs saw an odd number and did not test it

**More text, more to audit**

structured reviews were 60 to 80% longer; every point still needs checking

The structure came from the method you learned this morning and from the reporting guideline. Clean planted errors are an easy case: real manuscripts are harder, and a real manuscript under review never goes into an LLM.

# A safe routine for AI in your own appraisal work



# Wrap-up

**Open science**
compare the paper with what was promised

**Methodology**
estimand, interval, numbers, bias direction, spin

**Review**
problem, why it matters, what to do

**LLMs**
never confidential text; verify every claim; disclose

**Post-quiz**
Same 15 questions as the pre-quiz. Which answers changed, and which layer of the method explains the change?

Take-home kit: 7-layer checklist, toolkit, review report template. To go further: the reporting guideline for the designs you read (toolkit, layer 1).

## Post-quiz — individual

10

**minutes**

### START NOW

**Instructions and answer grid:** pack p. 26-27

Same 15 questions as the pre-quiz, one answer each.

**STOP when time is up → next: evaluation, and thank you**